

Mesh Convergence Study

David Puerta

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1 Overview

Using the power-law, force-driven LBM program, we will conduct a mesh convergence study for shear-thinning, shear-thickening and Newtonian fluids.

The mesh convergence study consists of running various, in our case 4, simulations of different mesh sizes with all the simulations having the same Reynolds number.

There is a relationship between each mesh size and number of iteration for that mesh. The starting mesh $M_1 = (N_x, N_y)$ is run for I_1 iterations. The following mesh $M_2 = (2N_x, 2N_y)$ is four times the size of M_1 and is run for $I_2 = 4I_1$ iterations. Following this reasoning, we have the n^{th} mesh $M_n = (nN_x, nN_y)$ is ran for $I_n = n^2I_1$ iterations. As mentioned, we will use $n = 4$ mesh sizes in this convergence study but it can be extended to more.

For this study we define the Reynolds number as,

$$\text{Re} = \frac{U_{max} N_y}{\nu_c} \quad (1)$$

where U_{max} is the maximum velocity of the fluid, N_y is the height of our mesh and ν_c is the apparent viscosity at the centreline which is found using

$$\nu_c = \frac{1}{3} \left(\tau_c - \frac{1}{2} \right) \quad (2)$$

where τ_c is the relaxation time at the centreline.

Moreover, since for consequent mesh sizes the size of N_y is doubled, we impose that the maximum fluid velocity U_{max} must be *halved*. Since our simulation takes inputs of n, m, g , we will find the value of g such that the maximum velocity is halved when our height N_y is doubled.

Recall that the maximum velocity U_{max} has the expression,

$$U_{max} = \frac{n}{n+1} \left(\frac{\rho g}{m} \right)^{1/n} \left(\frac{N_y}{2} \right)^{(n+1)/n} \quad (3)$$

from the analytic solution of a power-law fluid with force-driven flow $\mathbf{g} = (g, 0)$.

Rearranging this expression for g ,

$$g = \frac{2m}{\rho N_y} \left(\frac{2(n+1)U_{max}}{nN_y} \right)^n \quad (4)$$

Given a Reynolds number Re , we can find the kinematic viscosity by rearranging (1),

$$\nu_c = \frac{U_{max} N_y}{\text{Re}} \quad (5)$$

However, this presents an issue. In order to run the simulation, we must enter a value for the consistency coefficient m , yet we do not know which value of m will result in a Reynolds number Re - since the values of m and g change Re .

Workaround Using the analytic solution for the relaxation time¹,

$$\tau(y) = \frac{1}{2} + \frac{3m}{\rho} \left(\frac{\rho g}{m} \right)^{(n-1)/n} \left| \frac{N_y}{2} - y \right|^{(n-1)/n} \quad (6)$$

we can evaluate this expression near the centreline² at $y = \frac{N_y}{2} - \frac{1}{2}$,

$$\tau \left(\frac{N_y}{2} - \frac{1}{2} \right) = \tau_c = \frac{1}{2} + \frac{3m}{\rho} \left(\frac{\rho g}{2m} \right)^{(n-1)/n} \quad (7)$$

Using this, we can find an expression for the kinematic velocity at the centreline³,

$$\nu_c = \frac{m}{\rho} \left(\frac{\rho g}{2m} \right)^{(n-1)/n} \quad (8)$$

Now that we have all the ingredients, we have the following procedure to find the values of g and m for each mesh size:

1. Choose an initial maximum velocity U_{max} for the first mesh size.
2. Use (4) to find the relationship for g in terms of m .
3. Choose a Reynolds number Re and use (5) to find the corresponding kinematic viscosity at the centreline.
4. Using (8), solve for m and g .

We can now choose a Reynolds number and produce four mesh sizes with corresponding iterations and vary g and m for each mesh so that we have the same Re in each simulation. For Newtonian fluids, there is a further simplification as $\nu = m$ is constant and we need only vary g .

Once the simulations are ran, we can explore the errors. The LBM is of second order accuracy, for each simulation with corresponding mesh size, we calculate the error ϵ ,

$$\epsilon = \sqrt{\frac{\sum_{k=1}^N ||\mathbf{u}^{num}(k) - \mathbf{u}^{ana}(k)||^2}{\sum_{k=1}^N ||\mathbf{u}^{ana}(k)||^2}} \quad (9)$$

with the L^2 norm $||\cdot||$, analytic velocity field $\mathbf{u}^{ana}$, numerical velocity $\mathbf{u}^{num}$, site k , number of lattices $N = N_x \times N_y$ and plot $\log(\epsilon)$ on the y -axis and $\log(N)$ on the x -axis. Since we have second order accuracy for the LBM, we *expect* a straight line with gradient of -1 .

¹This comes from using the equation $\tau(y) = \frac{1}{2} + \frac{3}{\rho} \mu_{app}$ and $\mu_{app} = m \left| \frac{du_x}{dy} \right|^{n-1}$ together with the analytic solution for the velocity $u_x(y)$.

²We do not evaluate at the centreline since we would get $\tau = 1/2$, which gives us an issue with LBM.

³We hereby omit ‘near the centreline’ in favour of ‘at the centreline’ for convenience.

2 Very Small Reynolds Number

Here, we perform a mesh convergence study for power-law fluids with flow behaviour indices of $n = 0.5$, 1.0 , and 2.0 , representing shear-thinning, Newtonian, and shear-thickening behaviour, respectively, at a very small Reynolds number ($\text{Re} = 0.1$).

2.1 Shear-Thinning Fluid

For a shear-thinning fluid with $n = 0.5$, we have the following results, all with an analytic centreline relaxation time of $\tau_c = 0.740$.

Mesh	N_y	Iterations	U_{max}	g	m	$\log(\epsilon)$	$\log(N)$
M_1	8	25,000	1.00×10^{-3}	1.88×10^{-6}	2.74×10^{-4}	-3.22	4.85
M_2	16	100,000	5.00×10^{-4}	1.17×10^{-7}	6.85×10^{-5}	-4.60	6.24
M_3	32	400,000	2.50×10^{-4}	7.32×10^{-9}	1.71×10^{-5}	-5.74	7.62
M_4	64	1,600,000	1.25×10^{-4}	4.58×10^{-10}	4.28×10^{-6}	-3.68	9.01

Table 1: Simulation parameters and results for $\text{Re} = 0.1$, $n = 0.5$.

2.2 Newtonian Fluid

For a Newtonian fluid with $n = 1.0$, the consistency coefficient m is precisely the kinematic viscosity at the centreline ν_c . Since we are testing a Newtonian fluid, the relaxation time is constant. Here we have $\nu_c = 0.8$ from (1).

Mesh	N_y	Iterations	U_{max}	g	m	$\log(\epsilon)$	$\log(N)$
M_1	8	25,000	1.00×10^{-3}	1.00×10^{-5}	8.00×10^{-2}	-4.21	4.85
M_2	16	100,000	5.00×10^{-4}	1.25×10^{-6}	8.00×10^{-2}	-5.60	6.24
M_3	32	400,000	2.50×10^{-4}	1.56×10^{-7}	8.00×10^{-2}	-6.98	7.62
M_4	64	1,600,000	1.25×10^{-4}	1.95×10^{-8}	8.00×10^{-2}	-8.37	9.01

Table 2: Simulation parameters and results for $\text{Re} = 0.1$, $n = 1.0$.

2.3 Shear-Thickening Fluid

For a shear-thickening fluid with $n = 2.0$, we have the following results, all with a centreline relaxation time of $\tau_c = 0.740$.

Mesh	N_y	Iterations	U_{max}	g	m	$\log(\epsilon)$	$\log(N)$
M_1	8	25,000	1.00×10^{-3}	2.12×10^{-5}	6.03×10^2	-3.87	4.85
M_2	16	100,000	5.00×10^{-4}	3.75×10^{-6}	3.41×10^3	-4.37	6.24
M_3	32	400,000	2.50×10^{-4}	6.63×10^{-7}	1.93×10^4	-4.97	7.62
M_4	64	1,600,000	1.25×10^{-4}	1.17×10^{-7}	1.09×10^5	-5.62	9.01

Table 3: Simulation parameters and results for $\text{Re} = 0.1$, $n = 2.0$.

2.4 Discussion

Following the simulations, we plot the log error on the y -axis and the log of the mesh size on the x -axis. We then find the gradient of the straight line of best fit (SLOBF) and compare it to the hypothesised gradient of -1 .

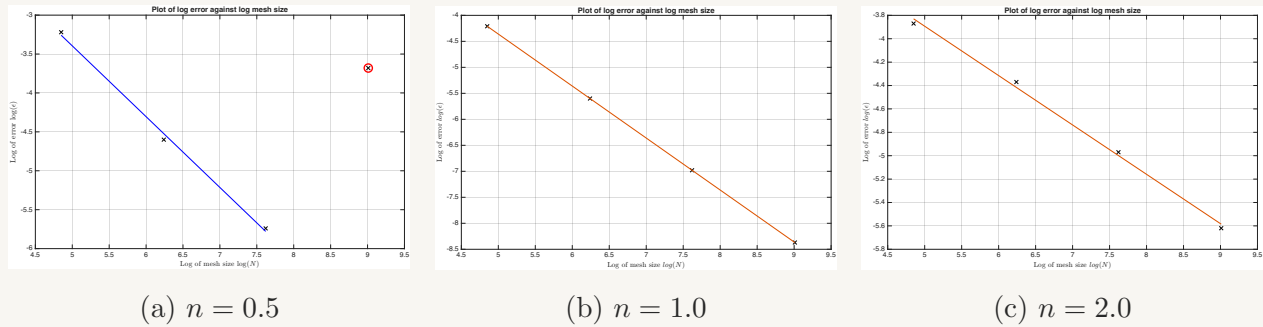


Figure 1: Plots for the fluids at $\text{Re} = 0.1$.

Flow behaviour index	Gradient of SLOBF	Percentage error
$n = 0.5$	-0.9098	9.02%
$n = 1.0$	-1.0000	0%
$n = 2.0$	-0.4221	57.79%

Table 4: Results for $\text{Re} = 0.1$.

The outlier for $n = 0.5$ is due to the minimum relaxation time not converging at the centreline, see *Appendix RT Mesh 4*. This could be due to the program's Newton-Raphson iteration being hardcoded to 14 iterations. The Newtonian fluid behaved as expected. Finally, for

$n = 2.0$, the L^2 error is quite high yet the data closely matches the analytic solution, see *Appendix Mesh 1-4*. This could be due to the very large values for the fluid consistency coefficient being very large, see *Table 4*.

3 Moderate Reynolds Number

Here, we perform a mesh convergence study for power-law fluids with flow behaviour indices of $n = 0.5$, 1.0 , and 2.0 , representing shear-thinning, Newtonian, and shear-thickening behaviour, respectively, at a moderate Reynolds number ($\text{Re} = 10.0$).

3.1 Shear-Thinning Fluid

For a shear-thinning fluid with $n = 0.5$, we have the following results, all with a centreline relaxation time of $\tau_c = 0.740$.

Mesh	N_y	Iterations	U_{max}	g	m	$\log(\epsilon)$	$\log(N)$
M_1	8	25,000	1.00×10^{-1}	1.88×10^{-4}	2.74×10^{-3}	-3.22	4.85
M_2	16	100,000	5.00×10^{-2}	1.17×10^{-5}	6.85×10^{-4}	-4.59	6.24
M_3	32	400,000	2.50×10^{-2}	7.32×10^{-8}	1.71×10^{-4}	-5.74	7.62
M_4	64	1,600,000	1.25×10^{-3}	4.58×10^{-8}	4.28×10^{-5}	-3.68	9.01

Table 5: Simulation parameters and results for $\text{Re} = 10.0$, $n = 0.5$.

3.2 Newtonian Fluid

For a Newtonian fluid with $n = 1.0$, the consistency coefficient m is precisely the kinematic viscosity at the centreline ν_c . Since we are testing a Newtonian fluid, the relaxation time is constant. Here we have $\nu_c = 0.08$ from (1).

Mesh	N_y	Iterations	U_{max}	g	m	$\log(\epsilon)$	$\log(N)$
M_1	8	25,000	1.00×10^{-1}	1.00×10^{-3}	8.00×10^{-2}	-4.21	4.85
M_2	16	100,000	5.00×10^{-2}	1.25×10^{-4}	8.00×10^{-2}	-5.60	6.24
M_3	32	400,000	2.50×10^{-2}	1.56×10^{-5}	8.00×10^{-2}	-6.98	7.62
M_4	64	1,600,000	1.25×10^{-2}	1.95×10^{-6}	8.00×10^{-2}	-8.37	9.01

Table 6: Simulation parameters and results for $\text{Re} = 10.0$, $n = 1.0$.

3.3 Shear-Thickening Fluid

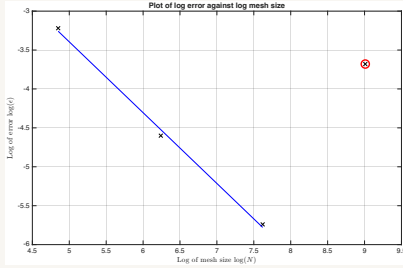
For a shear-thickening fluid with $n = 2.0$, we have the following results, all with a centreline relaxation time of $\tau_c = 0.740$.

Mesh	N_y	Iterations	U_{max}	g	m	$\log(\epsilon)$	$\log(N)$
M_1	8	25,000	1.00×10^{-1}	2.12×10^{-3}	6.03	-3.87	4.85
M_2	16	100,000	5.00×10^{-2}	3.75×10^{-4}	3.41×10^1	-4.40	6.24
M_3	32	400,000	2.50×10^{-2}	6.63×10^{-5}	1.93×10^2	-5.00	7.62
M_4	64	1,600,000	1.25×10^{-3}	1.17×10^{-5}	1.09×10^3	-5.62	9.01

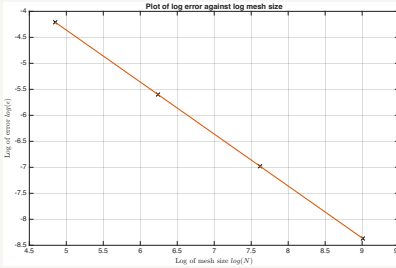
Table 7: Simulation parameters and results for $\text{Re} = 10.0$, $n = 2.0$.

3.4 Discussion

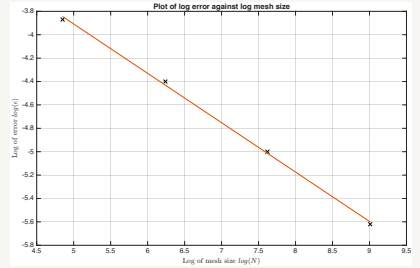
Following the simulations, we plot the log error on the y -axis and the log of the mesh size on the x -axis. We then find the gradient of the straight line of best fit (SLOBF) and compare it to the hypothesised gradient of -1 .



(a) $n = 0.5$



(b) $n = 1.0$



(c) $n = 2.0$

Figure 2: Plots for the fluids at $\text{Re} = 10.0$.

Flow behaviour index	Gradient of SLOBF	Percentage error
$n = 0.5$	-0.9098	9.02%
$n = 1.0$	-1.0000	0%
$n = 2.0$	-0.4221	57.79%

Table 8: Results for $\text{Re} = 10.0$.

Similar to the very small Reynolds case, we had an outlier for $n = 0.5$, Newtonian behaved as expected and $n = 2.0$ had a larger L^2 error than expected despite the data matching closely to the analytic solution.

4 Large Reynolds Number

Here, we perform a mesh convergence study for power-law fluids with flow behaviour indices of $n = 0.5$, 1.0 , and 2.0 , representing shear-thinning, Newtonian, and shear-thickening behaviour, respectively, at a large Reynolds number ($\text{Re} = 100$).

4.1 Shear-Thinning Fluid

For a shear-thinning fluid with $n = 0.5$, we have the following results, all with a centreline relaxation time of $\tau_c =$.

Mesh	N_y	Iterations	U_{max}	g	m	$\log(\epsilon)$	$\log(N)$
M_1	8	25,000	1.00	1.88×10^{-3}	8.66×10^{-3}	-3.23	4.85
M_2	16	100,000	5.00×10^{-1}	1.17×10^{-4}	2.17×10^{-3}	-4.60	6.24
M_3	32	400,000	2.50×10^{-1}	7.32×10^{-6}	5.41×10^{-4}	-5.74	7.62
M_4	64	1,600,000	1.25×10^{-1}	4.58×10^{-7}	1.35×10^{-4}	-3.68	9.01

Table 9: Simulation parameters and results for $\text{Re} = 100$, $n = 0.5$.

4.2 Newtonian Fluid

For a Newtonian fluid with $n = 1.0$, the consistency coefficient m is precisely the kinematic viscosity at the centreline ν_c . Since we are testing a Newtonian fluid, the relaxation time is constant. Here we have $\nu_c = 0.08$ from (1).

Mesh	N_y	Iterations	U_{max}	g	m	$\log(\epsilon)$	$\log(N)$
M_1	8	25,000	1.00	1.00×10^{-2}	8.00×10^{-2}	-4.21	4.85
M_2	16	100,000	5.00×10^{-1}	1.25×10^{-3}	8.00×10^{-2}	-5.60	6.24
M_3	32	400,000	2.50×10^{-1}	1.56×10^{-3}	8.00×10^{-2}	-6.98	7.62
M_4	64	1,600,000	1.25×10^{-1}	1.95×10^{-5}	8.00×10^{-2}	-8.37	9.01

Table 10: Simulation parameters and results for $\text{Re} = 100$, $n = 1.0$.

4.3 Shear-Thickening Fluid

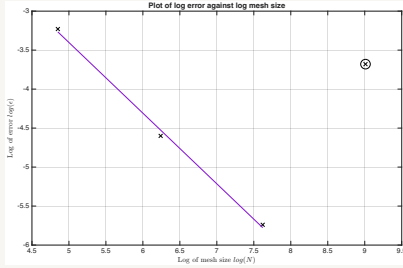
For a shear-thickening fluid with $n = 2.0$, we have the following results, all with a centreline relaxation time of $\tau_c =$.

Mesh	N_y	Iterations	U_{max}	g	m	$\log(\epsilon)$	$\log(N)$
M_1	8	25,000	1.00	2.12×10^{-2}	6.03×10^{-1}	-4.02	4.85
M_2	16	100,000	5.00×10^{-1}	3.75×10^{-3}	3.41	-4.46	6.24
M_3	32	400,000	2.50×10^{-1}	6.63×10^{-4}	1.93×10^1	-5.00	7.62
M_4	64	1,600,000	1.25×10^{-1}	1.17×10^{-4}	1.09×10^2	-5.63	9.01

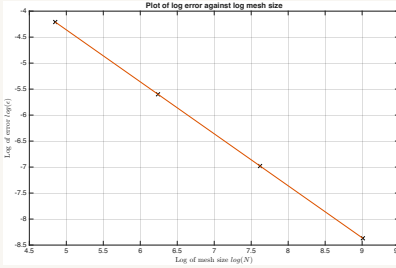
Table 11: Simulation parameters and results for $\text{Re} = 100$, $n = 2.0$.

4.4 Discussion

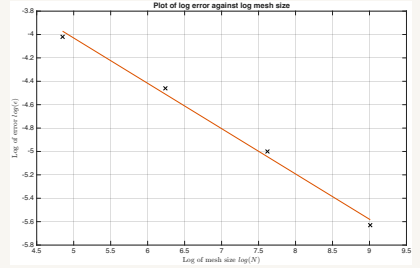
Following the simulations, we plot the log error on the y -axis and the log of the mesh size on the x -axis. We then find the gradient of the straight line of best fit (SLOBF) and compare it to the hypothesised gradient of -1 .



(a) $n = 0.5$



(b) $n = 1.0$



(c) $n = 2.0$

Figure 3: Plots for the fluids at $\text{Re} = 100$.

Flow behaviour index	Gradient of SLOBF	Percentage error
$n = 0.5$	-0.9062	9.38%
$n = 1.0$	-1.0000	0%
$n = 2.0$	-0.3874	61.26%

Table 12: Results for $\text{Re} = 100$.